KSP Datathon 2026 — Challenge 1

Intelligent Conversational AI & Crime Analytics Platform

SEMANTIC LAYER (C5)

Design Reference & Content Specification

Component	C5 — Domain Semantic Layer
Purpose	Translates natural-language queries into precise database concepts — crime types, BNS sections, metrics, schema mappings, and role-scoped filters
Sub-components	5 artifacts: Crime Type & Section Mapping · Domain Vocabulary Glossary · Metric Definitions · Schema-to-Concept Map · Role & Jurisdiction Resolver
Depends on	C4 (Query Orchestrator) · C6 (Text-to-SQL) · C15 (RBAC Policy Engine)
Feeds into	C4 (intent classification) · C6 (SQL generation) · C7 (RAG query construction)
Implementation	Static YAML/JSON files + dynamic context injection per query + role context injected every query
Status	Content designed — YAML encoding and system prompt template are next build steps
Document version	v1.0

1. What the Semantic Layer Is

C5 sits between the user's natural-language query and the query engines (C6 SQL, C7 RAG). It is a structured knowledge artifact — not code, not a prompt, not a database table — that tells the LLM what words mean in the KSP context before it tries to generate a query.

The semantic layer has four distinct jobs:

Job	Description
Translation	'Chain snatching' → crime_type_id = THEFT-CHAIN → primary_bns_section = BNS-309. Without this mapping the LLM guesses at column values and gets them wrong.
Disambiguation	'Detection rate' means something specific to KSP (FIRs where accused arrested or chargesheet filed, divided by total FIRs). Same with 'rowdy', 'history sheeter', 'bandobast', 'panchnama' — the LLM's generic understanding of these terms is either wrong or imprecise.
Jurisdiction resolution	When SP Meena asks about 'my district', the semantic layer knows her role maps to district_id = MYS. When an IO asks about 'my cases', it maps to io_officer_id = [her officer_id]. This is how RBAC and conversational context meet.
Schema grounding	Maps natural-language concepts to exact table names, column names, and valid enum values. 'Pending cases' → investigation_status IN ('Registered', 'Under_Investigation'). 'Absconding accused' → fir_accused.role = 'Absconder' OR arrests.is_absconding = TRUE.

Why C5 Is the Competitive Moat

Every team in this datathon will wire an LLM to a database. The semantic layer is what separates a credible domain-specific system from a generic demo. Research on text-to-SQL systems (Snowflake Cortex Analyst, Databricks Genie, ThoughtSpot) consistently shows that the semantic model — not the LLM — determines accuracy. On well-modelled data, text-to-SQL accuracy reaches ~95%; on unmapped data, the same LLM drops to ~40–60%.

The content that follows — BNS/IPC mappings, Kannada vocabulary, metric formulas, schema wiring, role filters — is the artifact that makes this system believable to a jury of serving police officers. A wrong BNS section number or an incorrect 'detection rate' formula in front of an SP will cost more credibility than any architectural gap.

2. Five Sub-Components at a Glance

Sub-component	Content & primary use
SC1 — Crime Type & Section Mapping	BNS↔IPC section table · crime type taxonomy · natural language → crime_type_id resolution. Used by almost every query.
SC2 — Domain Vocabulary Glossary	Bilingual (EN/KN) glossary of police procedure terms, investigation status values, and their precise database mappings. Prevents the LLM from guessing what 'rowdy' or 'chargesheet' means in schema terms.
SC3 — Metric Definitions	Precise formulas for detection rate, pendency rate, disposal rate, conviction rate, and all other KSP/NCRB metrics. Supervisor and policymaker queries depend entirely on this.
SC4 — Schema-to-Concept Map	Explicit wiring: entity concepts → tables → key columns → common joins. Time expression patterns. Implicit filter rules (the dangerous cases where a missing WHERE clause produces a wrong answer).
SC5 — Role & Jurisdiction Resolver	Per-role data scope, PII visibility, and aggregate-vs-records access. The cross-district antecedent exception. Merge point between C5 and C15 (RBAC).

3. SC1 — Crime Type & Section Mapping

This is the most referenced artifact — virtually every query about cases touches it. It has two layers: the BNS↔IPC section reference table (what sections mean) and the crime type taxonomy (what user words map to which sections).

BNS ↔ IPC Section Reference — Common Offences

NOTE *All FIRs registered before July 1, 2024 use IPC sections. All FIRs from July 1, 2024 onwards use BNS sections. The firs table stores both: primary_bns_section for post-BNS FIRs and ipc_sections[] for pre-BNS FIRs. The semantic layer must handle queries that span both eras.*

BNS Section	BNS Offence	IPC Equivalent	Category	Cogn.	Bailable	CS Days	Notes
`BNS 101`	Murder	IPC 302	Person	Yes	No	90	Most serious — 90-day chargesheet deadline
`BNS 103`	Culpable homicide not amounting to murder	IPC 304	Person	Yes	No	90	
`BNS 109`	Abetment of suicide	IPC 306	Person	Yes	No	60	
`BNS 115`	Voluntarily causing hurt	IPC 323	Person	Yes	Yes	60	Most common assault section
`BNS 117`	Voluntarily causing grievous hurt	IPC 325	Person	Yes	No	60	
`BNS 118`	Grievous hurt — dangerous weapons	IPC 326	Person	Yes	No	90	
`BNS 130`	Kidnapping	IPC 363	Person	Yes	No	60	
`BNS 137`	Kidnapping for ransom	IPC 364A	Person	Yes	No	90	
`BNS 63`	Rape	IPC 376	Sexual	Yes	No	90	POCSO applies if victim is minor
`BNS 74`	Assault to outrage modesty	IPC 354	Sexual	Yes	No	60	Often called 'molestation' colloquially
`BNS 85`	Cruelty by husband or relatives	IPC 498A	Person	Yes	No	60	Dowry harassment — very common
`BNS 303`	Theft	IPC 379	Property	Yes	Yes	60	Base theft — vehicle, mobile, etc.
`BNS 304`	Theft in dwelling house	IPC 380	Property	Yes	No	60	
`BNS 305`	Theft after preparation — house breaking	IPC 457	Property	Yes	No	60	Most common house-breaking section
`BNS 308`	Extortion	IPC 384	Property	Yes	No	60	
`BNS 309`	Robbery	IPC 392	Property	Yes	No	60	Chain snatching is commonly charged under this
`BNS 310`	Dacoity	IPC 395	Property	Yes	No	90	5+ persons — gang robbery
`BNS 317`	Receiving stolen property	IPC 411	Property	Yes	Yes	60	Receiver/fence — linked to theft cases
`BNS 318`	Cheating	IPC 420	Economic	Yes	No	60	'420 case' in common parlance
`BNS 319`	Cheating by impersonation	IPC 419	Economic	Yes	No	60	
`BNS 324`	Mischief — property damage	IPC 427	Property	Yes	Yes	60	

BNS Section	BNS Offence	IPC Equivalent	Category	Cogn.	Bailable	CS Days	Notes
`BNS 329`	Criminal breach of trust	IPC 406	Economic	Yes	No	60	
`BNS 331`	Dishonest misappropriation	IPC 403	Economic	Yes	Yes	60	
`BNS 238`	Counterfeiting currency	IPC 489A	Economic	Yes	No	90	
`BNS 351`	Criminal intimidation	IPC 506	Public Order	Yes	Yes	60	
`BNS 352`	Intentional insult	IPC 504	Public Order	No	Yes	—	Non-cognizable — no FIR, NCR only
`BNS 191`	Unlawful assembly / riot	IPC 147	Public Order	Yes	Yes	60	

Special Acts — No BNS Equivalent

NOTE *These acts continue to be cited by their original act and section number. The semantic layer must map both colloquial names and formal references to the correct act string.*

Act & Section	Offence	Category	CS Days	Common Alias
`NDPS Act S.20`	Possession / sale of cannabis	Narcotics	60	Ganja case, NDPS case
`NDPS Act S.21`	Possession / sale of manufactured drugs	Narcotics	60	Brown sugar, methamphetamine case
`NDPS Act S.29`	Abetment — NDPS offence	Narcotics	60	
`IT Act S.66C`	Identity theft	Cyber	60	Cyber crime, SIM fraud
`IT Act S.66D`	Cheating by impersonation — computer resource	Cyber	60	Online fraud, phishing
`IT Act S.67`	Publishing obscene material electronically	Cyber	60	
`SC/ST (PoA) Act S.3`	Atrocities against SC/ST members	Person	60	Atrocity case, SC/ST case
`Arms Act S.25`	Illegal possession of arms	Public Order	60	Arms case
`POCSO Act S.4`	Penetrative sexual assault on child	Sexual	90	POCSO case, child sexual abuse
`POCSO Act S.8`	Sexual assault on child	Sexual	60	
`Karnataka Excise Act`	Illicit liquor — manufacture/sale/possession	Public Order	60	Excise case, hooch case, TASMAC
`IPC 370 / BNS 143`	Human trafficking	Person	90	Trafficking case

Crime Type Taxonomy — Natural Language to crime_type_id

NOTE *This is the primary lookup table for query translation. When a user says 'chain snatching cases', the orchestrator finds THEFT-CHAIN here, then uses primary_bns_section = BNS-309 to filter the firs table. The Kannada column is used for voice and Kannada-script queries.*

Natural Language (English)	Kannada / Aliases	crime_type_id	Primary BNS	Notes
`chain snatching, chain theft, snatching`	ಸರಗಳ್ಳತನ, ಕೊರಳ ಸರ ಕಳ್ಳತನ	THEFT-CHAIN	BNS 309	Often on two-wheeler; also BNS 303 if no threat
`vehicle theft, bike theft, car theft, two-wheeler theft`	ವಾಹನ ಕಳ್ಳತನ, ಬೈಕ್ ಕಳ್ಳತನ	THEFT-VEHICLE	BNS 303	Use vehicles table for registration lookup
`house breaking, house burglary, break-in, B&E`	ಮನೆ ಕಳ್ಳತನ, ಮನೆ ಒಡೆದು ಕಳ್ಳತನ	THEFT-HOUSEBREAK	BNS 305	BNS 304 if no forced entry
`robbery, armed robbery, snatching with threat`	ದರೋಡೆ, ಲೂಟಿ	ROBBERY	BNS 309	BNS 310 for dacoity (5+ persons)

Natural Language (English)	Kannada / Aliases	crime_type_id	Primary BNS	Notes
mobile theft, phone theft, mobile snatching	ಮೊಬೈಲ್ ಕಳ್ಳತನ	THEFT-MOBILE	BNS 303	IMEI in serial_number field
jewellery theft, gold theft, ornament theft	ಆಭರಣ ಕಳ್ಳತನ, ಚಿನ್ನ ಕಳ್ಳತನ	THEFT-JEWELLERY	BNS 303	Often co-occurs with THEFT-HOUSEBREAK
murder, homicide, killing, death case	ಕೊಲೆ, ಹತ್ಯೆ	HOMICIDE	BNS 101	BNS 103 for culpable homicide
assault, attack, beating, hurt, fight	ಹಲ್ಲೆ, ಮಾರಾಮಾರಿ, ಹೊಡೆದಾಟ	ASSAULT	BNS 115	BNS 117/118 for grievous hurt
grievous hurt, serious assault, weapon attack	ತೀವ್ರ ಗಾಯ, ಆಯುಧ ಹಲ್ಲೆ	ASSAULT-GRIEVOUS	BNS 117	
rape, sexual assault	ಅತ್ಯಾಚಾರ	SEXUAL-ASSAULT	BNS 63	POCSO if victim < 18
molestation, eve teasing, harassment, outrage modesty	ಲೈಂಗಿಕ ಕಿರುಕುಳ	SEXUAL-MOLESTATION	BNS 74	
kidnapping, abduction	ಅಪಹರಣ	KIDNAPPING	BNS 130	BNS 137 if ransom demanded
cheating, fraud, 420 case, misrepresentation	ವಂಚನೆ, ಮೋಸ, ಮೋಸದ ಪ್ರಕರಣ	FRAUD-CHEATING	BNS 318	'420 case' = BNS 318
cyber crime, online fraud, phishing, OTP fraud, UPI fraud	ಸೈಬರ್ ಅಪರಾಧ, ಆನ್‌ಲೈನ್ ವಂಚನೆ	CYBER-FRAUD	IT 66D	
drug case, NDPS, narcotics, ganja, charas	ಮಾದಕ ದ್ರವ್ಯ, ಗಾಂಜಾ	NARCOTICS	NDPS 20	NDPS 21 for hard drugs
dowry harassment, 498A, domestic violence	ವರದಕ್ಷಿಣೆ ಕಿರುಕುಳ, ಗೃಹ ಹಿಂಸೆ	DOWRY-HARASSMENT	BNS 85	
missing person, missing child	ನಾಪತ್ತೆ, ಕಾಣೆಯಾದ ವ್ಯಕ್ತಿ	MISSING-PERSON	—	Not a cognizable crime in itself
riot, mob violence, unlawful assembly, communal	ಗಲಭೆ, ದಂಗೆ	PUBLIC-ORDER-RIOT	BNS 191	
arms case, illegal weapons, countrymade	ಅಕ್ರಮ ಶಸ್ತ್ರಾಸ್ತ್ರ	ARMS-ILLEGAL	Arms 25	
extortion, threatening, blackmail	ಸುಲಿಗೆ, ಬ್ಲ್ಯಾಕ್‌ಮೇಲ್	EXTORTION	BNS 308	
accident, road accident, MV Act, rash driving	ಅಪಘಾತ, ರಾಷ್ ಡ್ರೈವಿಂಗ್	ACCIDENT-MV	MV Act	Not a BNS section
atrocity case, SC/ST case	ದೌರ್ಜನ್ಯ ಪ್ರಕರಣ	ATROCITY-SCST	SC/ST 3	
excise case, hooch, illicit liquor	ಅಬಕಾರಿ ಪ್ರಕರಣ, ಕಳ್ಳಭಟ್ಟಿ	EXCISE	KE Act	Karnataka Excise Act
cheating by impersonation, impersonation	ಸೋಗು ಹಾಕಿ ಮೋಸ	FRAUD-IMPERSONATION	BNS 319	
property damage, mischief, vandalism	ಆಸ್ತಿ ನಾಶ, ದ್ವೇಷ ಕೃತ್ಯ	MISCHIEF	BNS 324	

4. SC2 — Domain Vocabulary Glossary (Bilingual)

These are the terms that appear in natural-language queries where a generic LLM either does not know the meaning or would produce an incorrect database mapping. Every term here has a precise schema consequence.

Police Procedure Terms

Term (EN)	Kannada	Definition for the LLM	Database Mapping
`FIR`	ಪ್ರಥಮ ಮಾಹಿತಿ ವರದಿ	First Information Report — the first step in a criminal case for a cognizable offence. Registered at the police station.	firs table — primary record
`NCR`	—	Non-Cognizable Report — complaint for a non-cognizable offence. No FIR, no arrest without warrant. Not in the main firs table.	Separate NCR register — not in v1 schema
`Case diary`	ಪ್ರಕರಣ ದಿನಚರಿ	Daily record of investigation maintained by the IO — legally mandated under BNSS. Contains the richest free-text investigative narrative.	case_diary_entries table
`Chargesheet`	ದೋಷಾರೋಪ ಪಟ್ಟಿ	Final report filed by police to the court after investigation. Also called 'Final Report'. Filing triggers the 60/90-day BNSS deadline.	chargesheets table
`Closure report`	ಮುಕ್ತಾಯ ವರದಿ	Filed when investigation is closed — True: crime occurred but accused untraced; False: no crime occurred.	firs.investigation_status = 'Closed_True' / 'Closed_False'
`Panchnama`	ಪಂಚನಾಮೆ	On-scene document witnessed by two independent persons during a search or seizure. Required for evidence admissibility.	Mentioned in case_diary_entries.entry_text
`Mahazar`	ಮಹಜರ್	Scene of offence inspection document — records physical state of the scene.	Mentioned in case_diary_entries.entry_text
`Remand`	ರಿಮಾಂಡ್	Judicial order authorising continued detention of the accused.	arrests.remand_type, arrests.remand_expiry
`PC remand`	ಪೊಲೀಸ್ ಕಸ್ಟಡಿ	Police Custody remand — accused in police custody for further interrogation. Maximum 15 days under BNSS.	arrests.remand_type = 'Police_Custody'
`JC remand`	ನ್ಯಾಯಾಂಗ ಬಂಧನ	Judicial Custody — accused sent to jail while investigation continues.	arrests.remand_type = 'Judicial_Custody'
`Absconding`	ತಲೆಮರೆಸಿಕೊಂಡ	Accused who has fled and cannot be located for arrest.	fir_accused.role = 'Absconder' OR arrests.is_absconding = TRUE
`Proclaimed offender`	ಘೋಷಿತ ಅಪರಾಧಿ	Court-declared absconder — evading arrest despite a warrant issued by the court.	persons.is_proclaimed_offender = TRUE
`Zero FIR`	ಜೀರೋ ಎಫ್ಐಆರ್	FIR registered at a non-jurisdictional station and then transferred to the correct station. Valid under BNSS.	firs.is_zero_fir = TRUE
`Suo motu`	ಸ್ವಯಂ ಪ್ರೇರಣೆ	FIR registered by police on their own initiative, without a complainant.	firs.fir_type = 'Suo_Motu'
`Bandobast`	ಬಂದೋಬಸ್ತ್	Deployment of police personnel for an event, VIP visit, election, or sensitive occasion.	events_calendar table; officers table (deployment)
`Beat`	ಬೀಟ್	Defined patrol area assigned to an officer or patrol vehicle (Hoysala).	officers context; referenced in station records
`History sheeter`	ಇತಿಹಾಸ ಶೀಟ್ ವ್ಯಕ್ತಿ	Person with an active history sheet — a repeat offender under police surveillance. Karnataka-specific procedure.	persons.is_history_sheeted = TRUE; history_sheets table
`Rowdy`	ರೌಡಿ	Informal term for a history-sheeted person in the Rowdy or Goonda category.	history_sheets.category IN ('Rowdy','Goonda')

Term (EN)	Kannada	Definition for the LLM	Database Mapping
Habitual offender	ಅಭ್ಯಾಸ ಅಪರಾಧಿ	Person with 3+ registered FIRs as accused — threshold for history sheet opening in Karnataka.	Computed: COUNT(fir_accused) >= 3 GROUP BY person_id
Modus operandi / MO	ಅಪರಾಧ ವಿಧಾನ	Method and pattern used by an offender to commit crimes. Stored as a code and as free text.	mo_codes table; firs.mo_code_id; firs.mo_description_free
FSL	ವೈಜ್ಞಾನಿಕ ಪ್ರಯೋಗಾಲಯ	Forensic Science Laboratory. IO sends physical evidence here for analysis.	Referenced in case_diary_entries.action_taken
Acquittal	ಖಾಲಾಸ	Court verdict — accused found not guilty.	court_disposals.outcome = 'Acquitted'
Conviction	ಶಿಕ್ಷೆ	Court verdict — accused found guilty.	court_disposals.outcome = 'Convicted'
Compounding	ರಾಜಿ	Settlement between complainant and accused — case withdrawn.	court_disposals.outcome = 'Compounded'

Investigation Status — Exhaustive Mapping

NOTE *This mapping is critical. 'Pending', 'open', 'solved', and 'closed' are used loosely in everyday language but have precise schema values. Using the wrong status filter produces large errors in case counts.*

Natural Language	Kannada	investigation_status values	Additional condition
Open / pending cases	ತೆರೆದ ಪ್ರಕರಣಗಳು	'Registered', 'Under_Investigation'	—
Solved / detected cases	ಪತ್ತೆಯಾದ ಪ್ರಕರಣಗಳು	'Chargesheet_Filed'	AND at least one arrest in fir_accused
Chargesheet filed	ದೋಷಾರೋಪ ಸಲ್ಲಿಕೆ	'Chargesheet_Filed'	—
Closed / disposed	ಮುಕ್ತಾಯ	'Closed_True', 'Closed_False', 'Referred'	—
Undetected	ಪತ್ತೆಯಾಗದ	'Closed_True'	Crime confirmed but accused not traced — count as undetected
False case / no crime	ಸುಳ್ಳು ಪ್ರಕರಣ	'Closed_False'	No crime found to have occurred
Pending trial	ವಿಚಾರಣೆ ಬಾಕಿ	'Chargesheet_Filed'	AND court_disposals is null OR outcome = 'Pending'
Convicted	ಶಿಕ್ಷೆಯಾಗಿದೆ	'Chargesheet_Filed'	AND court_disposals.outcome = 'Convicted'
Acquitted	ಖಾಲಾಸ ಆಗಿದೆ	'Chargesheet_Filed'	AND court_disposals.outcome = 'Acquitted'
Registered today / this week	ಇಂದು ದಾಖಲಾದ	Any status	Filter on registration_date — use time expression patterns in SC4

5. SC3 — Metric Definitions

These are the official KSP/NCRB metrics that supervisors and policymakers ask about. Every one has a precise formula — a generic LLM will approximate and get it wrong in a review meeting. These definitions are the single most important correctness check for the supervisor and policymaker personas.

> **NOTE** *All rate metrics must be computed within an explicit time period. An unqualified query ('what is the detection rate?') must prompt the system to ask which period is intended, or default to the current calendar year with a stated assumption in the response.*

Metric	Kannada	Precise Formula	Table / columns
`Detection rate`	ಪತ್ತೆ ದರ	(FIRs with at least one arrest OR chargesheet filed in period) ÷ (Total FIRs registered in same period) × 100. NCRB definition. Does NOT require conviction.	firs + fir_accused (is_arrested) + chargesheets. Filter by registration_date.
`Pendency rate`	ಬಾಕಿ ದರ	(FIRs in 'Registered' or 'Under_Investigation' at period end, across ALL years) ÷ (Total FIRs ever registered still open). Includes carry-forward from prior years.	firs WHERE investigation_status IN ('Registered','Under_Investigation'). No date filter on numerator.
`Disposal rate`	ವಿಲೇ ದರ	(FIRs disposed in period — chargesheet filed + closed) ÷ (FIRs registered in same period) × 100. Period-bounded; does NOT include carry-forward.	firs. Filter registration_date AND disposed_date within same period.
`Chargesheet rate`	ದೋಷಾರೋಪ ದರ	(Chargesheets filed in period) ÷ (FIRs registered in same period) × 100. Subset of disposal rate.	chargesheets.filing_date within period ÷ firs.registration_date within period.
`Conviction rate`	ಶಿಕ್ಷೆ ದರ	(Convicted disposals) ÷ (Total disposals with a recorded outcome) × 100. Operates on court_disposals, NOT firs.	court_disposals WHERE outcome != 'Pending'. Does not filter by FIR registration date.
`Cases registered`	ದಾಖಲಾದ ಪ್ರಕರಣಗಳು	COUNT(firs) WHERE registration_date BETWEEN [start] AND [end]. Always filter by period — unqualified count is meaningless.	firs. Always require date filter.
`Repeat offender`	ಪುನರಾವರ್ತಿತ ಅಪರಾಧಿ	Person appearing as accused in 2 or more distinct FIRs at any time.	COUNT(DISTINCT fir_id) >= 2 GROUP BY person_id in fir_accused.
`Habitual offender`	ಅಭ್ಯಾಸ ಅಪರಾಧಿ	Person with 3 or more distinct FIRs as accused — threshold for history sheet opening in Karnataka.	COUNT(DISTINCT fir_id) >= 3 GROUP BY person_id in fir_accused.
`Crime rate`	ಅಪರಾಧ ದರ	(FIRs registered in district in period) ÷ (District population) × 100,000.	firs JOIN district_socioeconomic ON district_id and data_year. Never compute without population join.
`Policing density`	ಪೊಲೀಸ್ ಸಾಂದ್ರತೆ	Officers per lakh population — pre-computed field; do not recalculate from the officers table directly.	district_socioeconomic.officers_per_lakh_pop.
`Arrest rate`	ಬಂಧನ ದರ	(Total arrests in period) ÷ (FIRs registered in period) × 100. One FIR may have multiple arrests.	arrests.arrest_date within period ÷ firs.registration_date within period.

6. SC4 — Schema-to-Concept Map

This sub-component wires natural-language concepts to exact database objects, defines how to resolve relative time expressions, and documents the implicit filter rules that must be applied even when the user does not state them.

Entity Concepts → Tables

User concept	Primary table	Key columns	Common joins needed
`A case / an FIR`	firs	fir_id, crime_type_id, investigation_status, registration_date, chargesheet_deadline	fir_accused (accused), fir_victims (victims), case_diary_entries (narrative), chargesheets (outcome)
`A suspect / accused`	fir_accused + persons	person_id, role, is_arrested, accused_serial	persons (identity), arrests (remand/bail), history_sheets (antecedents)
`A victim`	fir_victims + persons	person_id, victim_serial, injury_type	firs (case context). Never expose victim PII without case linkage.
`An officer / IO`	officers	officer_id, rank, role, station_id, district_id	firs.io_officer_id (their cases), firs.sho_officer_id (station cases)
`A station`	police_stations	station_id, station_name, circle_id, district_id, station_type	firs (cases registered there), officers (posted there)
`My district`	districts	district_id — RESOLVED from officers.district_id at query time for current user	firs, police_stations, district_socioeconomic
`My cases`	firs	Filtered: io_officer_id = [current officer_id]	Always apply IO scope unless role is Supervisor or above
`A gang`	gangs	gang_id, gang_name, primary_crime_type, operating_district, is_active	gang_memberships (members), persons (member identities)
`A history sheeter`	history_sheets	history_sheet_id, person_id, category, risk_level, total_cases_count	persons (identity), history_sheet_entries (notes), firs (linked cases)
`A vehicle`	vehicles	registration_number (primary lookup), is_stolen, vehicle_type, make, model	persons (owner), firs (stolen_fir_id)
`Stolen property`	stolen_property	property_type, serial_number, is_recovered, description	firs (source FIR), firs (recovery_fir_id)
`An event / festival`	events_calendar	event_name, event_type, event_date_start, district_id	firs (incidents during event window), districts
`Connected persons`	known_associates	person_id_a, person_id_b, association_type, confidence	persons (both ends), firs (first_seen_fir_id)
`Phone number linkage`	person_phones	phone_number, person_id, is_active	persons (owner). Use for cross-case phone matching.
`District indicators`	district_socioeconomic	population, literacy_rate, unemployment_rate, per_capita_income_inr	districts, firs (for crime rate computation)

Time Expression → SQL Pattern

NOTE *Relative time expressions must be resolved consistently. Default to the current financial year (April–March) for unqualified 'this year' queries — Karnataka government reporting follows the financial year, not calendar year.*

User says	SQL pattern (PostgreSQL)	Clarification needed?
`this month`	registration_date >= DATE_TRUNC('month', CURRENT_DATE) AND registration_date <= CURRENT_DATE	No
`last month`	registration_date >= DATE_TRUNC('month', CURRENT_DATE - INTERVAL '1 month') AND registration_date < DATE_TRUNC('month', CURRENT_DATE)	No
`this year`	registration_date >= DATE_TRUNC('year', CURRENT_DATE) AND registration_date <= CURRENT_DATE [OR financial year April–March]	Clarify calendar vs financial year if metric-related
`last year`	registration_date >= DATE_TRUNC('year', CURRENT_DATE - INTERVAL '1 year') AND registration_date < DATE_TRUNC('year', CURRENT_DATE)	No
`last 30 days`	registration_date >= CURRENT_DATE - INTERVAL '30 days'	No
`last quarter`	Previous 3-month block aligned to Apr/Jul/Oct/Jan	No
`last week`	registration_date >= DATE_TRUNC('week', CURRENT_DATE - INTERVAL '1 week') AND registration_date < DATE_TRUNC('week', CURRENT_DATE)	No
`recent (unqualified)`	Default: CURRENT_DATE - INTERVAL '30 days'. State assumption in response.	State assumption
`yesterday`	registration_date = CURRENT_DATE - INTERVAL '1 day'	No
`today`	registration_date = CURRENT_DATE	No
`this financial year`	registration_date >= [current April 1] AND registration_date <= CURRENT_DATE	No — Karnataka uses Apr–Mar
`2023-24`	registration_date >= '2023-04-01' AND registration_date < '2024-04-01'	No — financial year interpretation

Implicit Filter Rules

NOTE *These are the most dangerous omissions. A query that seems complete — 'show me all pending cases' — is dangerously underspecified without role-based scope. Every generated SQL query must be checked against these rules before execution.*

Situation	Implicit filter to add	Reason
`IO asks 'my cases'`	WHERE firs.io_officer_id = [current_officer_id]	An IO must never see another IO's cases by default.
`IO asks 'cases in my station'`	WHERE firs.station_id = [current_station_id]	Station jurisdiction scoping.
`SHO asks 'station cases'`	WHERE firs.station_id = [current_station_id]	SHO sees entire station, not just own cases.
`SP asks 'my district'`	WHERE firs.district_id = [current_district_id]	District jurisdiction scoping.
`Analyst asks 'all cases'`	No station filter — analyst has district/state scope depending on assignment	Analyst RBAC is broader than IO or SHO.
`Any role queries victims`	JOIN fir_victims ON fir_id — only for FIRs the user can access. Never expose victim name/address otherwise.	Privacy + RBAC. Victim PII is restricted by default.
`Policymaker queries`	Return aggregates only — no individual FIR IDs, no person names, no addresses	Policymakers see trends and summaries, never individual records.
`IO requests statewide antecedents`	Read access to fir_accused.person_id statewide — but response shows FIR header only (ID, section, station, status), not case diary or victim details from other stations	Legitimate cross-district exception — Story 1. Scoped response.
`Any query about a named private individual`	Log query in query_audit_log with records_accessed array. Flag if no case linkage exists between querying officer and the named person.	Audit requirement — Story 26 anomaly detection.

Situation	Implicit filter to add	Reason
`Unqualified 'all cases' without date filter`	Add WHERE registration_date >= CURRENT_DATE - INTERVAL '1 year' as default, state assumption in response	Prevents accidentally querying the entire historical dataset.

7. SC5 — Role & Jurisdiction Resolver

This sub-component defines what each role can see, at what granularity, and how the semantic layer and RBAC engine (C15) merge at query time. Every session has an authenticated officer — the resolver derives the correct scope filters from their role before any query is constructed.

Role Data Scope Matrix

Role	Data scope	Victim names / PII	Accused PII	Audit logs	Records or aggregates
`IO`	Own station + own assigned cases	Yes — linked cases only	Yes — linked cases only	Own queries only	Records (full detail for own cases)
`SHO`	Entire station	Yes — station cases only	Yes — station cases	Own queries only	Records (full station)
`Circle Inspector`	All stations in circle	Aggregated only	Yes	No	Records + aggregates
`Supervisor (SP/DCP)`	Entire district	No — aggregates only	Aggregated only	No	Aggregates + case summaries
`Analyst (DCRB)`	District or state (per assignment)	No	No	No	Aggregates only
`Policymaker`	State-wide	No	No	No	Aggregates only — no individual records
`Admin (SCRB)`	All data + full audit logs	Yes (audit purpose)	Yes (audit purpose)	Full access	All levels

Cross-District Antecedent Exception

An IO querying a suspect's statewide history (Story 1) is a legitimate exception to the station-scope rule. The resolver handles it as follows:

- Read access to fir_accused records statewide is permitted for antecedent lookup on a named person.
- The response is scoped: show FIR ID, crime type, section, station name, and investigation status only.
- The case diary, victim details, witness statements, and full FIR narrative from other stations are NOT returned.
- The query is logged with flag_reason = NULL but records_accessed array is fully populated for audit.
- If the querying IO has an existing case linkage to the suspect, full case details for that FIR are returned.

Session Context Payload

The following context is injected at the start of every session and used by the resolver to construct implicit filters:

```
{
  'officer_id':    'KSP-12345',
  'officer_name':  'Kavitha Nair',
  'rank':          'Inspector',
  'role':          'IO',
  'station_id':    'KA-BLR-047',
  'station_name':  'Indiranagar Police Station',
  'district_id':   'BLR',
  'district_name': 'Bengaluru Urban',
  'scope':         'station',       // station | district | state
  'pii_access':    'linked_only',   // linked_only | aggregated | none | full
  'record_access': 'records',       // records | aggregates | all
  'audit_scope':   'own'            // own | none | full
```

```
}
```

8. Implementation: How to Build C5

The semantic layer is not a database table and not a single prompt. It is three coordinated artifacts that work together at query time.

Part A — Static YAML Files (build once, version-control)

One YAML file per sub-component. These are content artifacts, not code. They should be reviewed for domain accuracy before any query testing begins.

File	Contents	Size estimate
`sc1_crime_types.yaml`	crime_type_id taxonomy, BNS↔IPC section reference, natural language aliases in EN and KN	~150 entries
`sc2_vocabulary.yaml`	Police procedure terms, investigation status mappings, bilingual glossary	~60 terms
`sc3_metrics.yaml`	Metric names, formulas, table references, SQL templates	~15 metrics
`sc4_schema_map.yaml`	Entity→table mappings, column references, join patterns, time expression SQL, implicit filter rules	~30 entity types + 12 time patterns + 10 implicit rules
`sc5_role_resolver.yaml`	Role scope matrix, PII access levels, session context payload schema, cross-district exception rules	~10 role definitions

Part B — Dynamic Context Injection (assembled per query by C4)

The full semantic layer is too large to inject into every query. The query orchestrator (C4) classifies the query first and pulls only the relevant sub-component sections. Classification takes one fast LLM call with a short system prompt and ~20 few-shot examples.

Query type (C4 classification)	Sub-components injected	Example query
`Crime lookup / count`	SC1 (crime taxonomy) + SC3 (metric if rate) + SC4 (time patterns + implicit filters)	'How many chain snatching cases in Mysuru last month?'
`Person / antecedent lookup`	SC2 (vocabulary: history sheet, rowdy) + SC4 (persons entity map + cross-district rule) + SC5 (role scope)	'Does Rajan Gowda have prior cases?'
`Network / association query`	SC4 (known_associates, gang entities) + SC2 (MO vocabulary) + SC5 (role scope)	'Who is connected to this accused?'
`MO / semantic search`	SC1 (crime taxonomy) + SC2 (MO vocabulary) → routes to C7 (RAG), not C6 (SQL)	'Find similar past cases — accused on two-wheeler...'
`Metric / trend query`	SC3 (metric definitions) + SC4 (time patterns) + SC5 (district scope for role)	'What is the detection rate in my district this quarter?'
`Deadline / compliance`	SC2 (chargesheet definition) + SC4 (chargesheet_deadline column) + SC5 (IO scope)	'Which of my cases are approaching the chargesheet deadline?'
`Event / bandobast query`	SC4 (events_calendar entity) + SC4 (time window pattern)	'What incidents happened during Dasara in Mysuru last year?'
`Socio-economic / overlay`	SC4 (district_socioeconomic entity) + SC3 (crime rate formula) + SC5 (policymaker scope)	'Correlate crime rate with unemployment in North Karnataka districts'

Part C — Role Context (injected every query, always)

The session context payload from SC5 is injected into every query prompt, regardless of query type. It is the merge point between C5 and C15 (RBAC). It is never omitted, never cached across sessions, and never overrideable by the user's query content.

NOTE *RBAC enforcement must happen at the data layer (WHERE clauses in generated SQL) — not at the prompt layer ('please don't show sensitive data'). A well-crafted query should not be able to bypass role scoping by phrasing the request cleverly.*

9. What to Build Next

The semantic layer content above is complete enough to start building. The four steps below are in strict order — each one gates the next.

Step	Task
Step 1 — Content (2–3 hrs)	Encode all five sub-components as YAML files. This is a content task, not a code task. Populate from the tables in this document. Review each file against the actual schema column names to catch any drift.
Step 2 — System prompt template (1–2 hrs)	Write the wrapper that takes static YAML content + dynamic role context and assembles the LLM system prompt. Decide injection format: XML tags (recommended for Claude models), markdown sections, or JSON schema.
Step 3 — Query classifier (2–3 hrs)	A lightweight first-pass prompt that reads a user query and outputs a classification label (crime_lookup / person_lookup / network / mo_search / metric / deadline / event / socioeconomic). Use 20 few-shot examples covering edge cases. This drives Part B injection.
Step 4 — Grounding validator (1–2 hrs)	A post-generation check that confirms every table name and column name in the generated SQL exists in the actual schema. This is the hallucination firewall for the query layer. Implement as a simple AST parse over the SQL string against a whitelist of valid identifiers.

— End of document —